

Lecture 1Mathematical LogicPropositional Calculus

\* Proposition is a statement which can ~~be~~ either be true or false.

\* Proposition is represented by small letters 'a', 'b', etc.

| $p$ | $q$ | $p \wedge q$ | $p \vee q$ | $\neg p$ | $\neg q$ |
|-----|-----|--------------|------------|----------|----------|
| T   | T   | T            | T          | F        | F        |
| T   | F   | F            | T          | F        | T        |
| F   | T   | F            | T          | T        | F        |
| F   | F   | F            | F          | T        | T        |

conjunction  $\rightarrow \wedge$  and  
disjunction  $\rightarrow \vee$  or  
negation  $\rightarrow \neg$

$$(*) \quad p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r)$$

| $p$ | $q$ | $r$ | $q \vee r$ | $p \wedge (q \vee r)$ | $p \wedge q$ | $p \wedge r$ | $(p \wedge q) \vee (p \wedge r)$ |
|-----|-----|-----|------------|-----------------------|--------------|--------------|----------------------------------|
| T   | T   | T   | T          | T                     | T            | T            | T                                |
| T   | T   | F   | T          | T                     | T            | F            | T                                |
| T   | F   | T   | T          | T                     | F            | T            | T                                |
| T   | F   | F   | F          | F                     | F            | F            | F                                |
| F   | T   | T   | T          | F                     | F            | F            | F                                |
| F   | T   | F   | T          | F                     | F            | F            | F                                |
| F   | F   | T   | T          | F                     | F            | F            | F                                |
| F   | F   | F   | F          | F                     | F            | F            | F                                |

Therefore,  $p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r)$

Lecture 2Q1

Consider the following

 $p$ : He is rich $q$ : He is generousSol 1:1.  $p \wedge q$  = He is rich and He is generous.2.  $p \vee q$  = He is rich or He is generous.3.  $\neg p$  = He is not rich.4.  $\neg q$  = He is not generous.5.  $\neg(\neg p)$  = It is false that He is not rich.6.  $(\neg p \wedge \neg q)$  = He is neither rich nor generous.7.  $\neg(p \vee q)$  = It is false that He is rich or He is generous.Q2 $p$ : It is hot day. $q$ : The temperature is  $45^\circ\text{C}$ .1.  $\neg p$  = It is not a hot day.2.  $p \wedge q$  = It is a hot day and the temperature is  $45^\circ\text{C}$ .3.  $\neg q$  = The temperature is not  $45^\circ\text{C}$ .4.  $p \vee q$  = It is a hot day or the temperature is  $45^\circ\text{C}$ .5.  $\neg p \wedge \neg q$  = Neither it is a hot day nor the temperature is  $45^\circ\text{C}$ .6.  $\neg(\neg q)$  = It is false that the temperature is not  $45^\circ\text{C}$ .7.  $\neg(p \wedge q)$  = It is false that it is a hot day and temp is  $45^\circ\text{C}$ .8.  $\neg(p \vee q)$  = It is false that it is a hot day or temp is  $45^\circ\text{C}$ .9.  $\neg(\neg p \vee \neg q)$  = It is false that neither it is a hot day nor the temperature is  $45^\circ\text{C}$ .



Derived ConnectivesNAND ( $\uparrow$ )

| P | q | $P \uparrow q$ |
|---|---|----------------|
| T | T | F              |
| F | F | T              |
| F | T | T              |
| T | F | T              |

NOR ( $\downarrow$ )

| P | q | $P \downarrow q$ |
|---|---|------------------|
| T | T | F                |
| T | F | F                |
| F | T | F                |
| F | F | T                |

XOR (Exclusive OR)

| P | q | $P \oplus q$ |
|---|---|--------------|
| T | T | F            |
| T | F | T            |
| F | T | T            |
| F | F | F            |

Q: Generate

1.  $p \oplus q \oplus r$

2.  $p \uparrow q \uparrow r$

| P | q | r | $P \oplus q$ | $P \oplus q \oplus r$ | $P \uparrow q$ | $P \uparrow q \uparrow r$ |
|---|---|---|--------------|-----------------------|----------------|---------------------------|
| T | T | T | F            | T                     | F              | T                         |
| T | T | F | F            | F                     | F              | T                         |
| T | F | T | T            | F                     | T              | F                         |
| T | F | F | T            | T                     | T              | T                         |
| F | T | T | T            | F                     | T              | F                         |
| F | T | F | T            | T                     | T              | T                         |
| F | F | T | F            | T                     | T              | F                         |
| F | F | F | F            | F                     | T              | T                         |



Q: Prove that

1.  $p \oplus q = (p \wedge \neg q) \vee (\neg p \wedge q)$

| p | q | $\neg p$ | $\neg q$ | $p \oplus q$ | $(p \wedge \neg q)$ | $(\neg p \wedge q)$ | $(p \wedge \neg q) \vee (\neg p \wedge q)$ |
|---|---|----------|----------|--------------|---------------------|---------------------|--------------------------------------------|
| T | T | F        | F        | F            | F                   | F                   | F                                          |
| T | F | F        | T        | T            | T                   | F                   | T                                          |
| F | T | T        | F        | T            | F                   | T                   | T                                          |
| F | F | T        | T        | F            | F                   | F                   | F                                          |

therefore,  $p \oplus q = (p \wedge \neg q) \vee (\neg p \wedge q)$ 

2.  $(p \oplus q) \vee (p \downarrow q) = p \uparrow q$

| p | q | $p \oplus q$ | $p \downarrow q$ | $p \uparrow q$ | $(p \oplus q) \vee (p \downarrow q)$ |
|---|---|--------------|------------------|----------------|--------------------------------------|
| T | T | F            | F                | F              | F                                    |
| T | F | T            | F                | T              | T                                    |
| F | T | T            | F                | T              | T                                    |
| F | F | F            | T                | T              | T                                    |

Therefore,  $(p \oplus q) \vee (p \downarrow q) = p \uparrow q$ 

3.  $(p \uparrow q) \oplus (p \uparrow q) = (p \vee q) \wedge (p \downarrow q)$

| p | q | $p \uparrow q$ | $p \uparrow q$ | $p \vee q$ | $p \downarrow q$ | $(p \uparrow q) \oplus (p \uparrow q)$ | $(p \vee q) \wedge (p \downarrow q)$ |
|---|---|----------------|----------------|------------|------------------|----------------------------------------|--------------------------------------|
| T | T | F              | F              | T          | F                | F                                      | F                                    |
| T | F | T              | T              | T          | F                | F                                      | F                                    |
| F | T | T              | T              | T          | F                | F                                      | F                                    |
| F | F | T              | T              | F          | T                | F                                      | F                                    |



Some other connectors1. Conditional ( $\rightarrow$ ) :-

| P | q | $P \rightarrow q$ |
|---|---|-------------------|
| T | T | T                 |
| T | F | F                 |
| F | T | T                 |
| F | F | T                 |

\* Contrapositive :-

The proposition  $\boxed{-q \rightarrow -p}$  is contrapositive of  $p \rightarrow q$ .

\* Converse :-

The proposition  $\boxed{q \rightarrow p}$  is called the converse of  $p \rightarrow q$ .

\* Inverse :-

The proposition  $\boxed{-p \rightarrow -q}$  is called the inverse of  $p \rightarrow q$ .

2. Bi-Conditional ( $\leftrightarrow$ ) :-

$$p \leftrightarrow q$$

p if and only if q

| P | q | $P \leftrightarrow q$ |
|---|---|-----------------------|
| T | T | T                     |
| T | F | F                     |
| F | T | F                     |
| F | F | T                     |

Q: Show that  $p \leftrightarrow q = (p \rightarrow q) \wedge (q \rightarrow p)$



| P | q | $p \leftrightarrow q$ | $p \rightarrow q$ | $q \rightarrow p$ | $(p \rightarrow q) \wedge (q \rightarrow p)$ |
|---|---|-----------------------|-------------------|-------------------|----------------------------------------------|
| T | T | T                     | T                 | T                 | T                                            |
| T | F | F                     | F                 | T                 | F                                            |
| F | T | F                     | T                 | F                 | F                                            |
| F | F | T                     | T                 | T                 | T                                            |

Therefore,  $p \leftrightarrow q = (p \rightarrow q) \wedge (q \rightarrow p)$

### \* Tautologies Tautologies (All true) :-

If a statement gives all values true in the final column it is called a tautology.

Q: Show that  $(p \rightarrow q) \leftrightarrow (-q \rightarrow -p)$  is a tautology

| P | q | $p \rightarrow q$ | $-q$ | $-p$ | $-q \rightarrow -p$ | $(p \rightarrow q) \leftrightarrow (-q \rightarrow -p)$ |
|---|---|-------------------|------|------|---------------------|---------------------------------------------------------|
| T | T | T                 | F    | F    | T                   | T                                                       |
| T | F | F                 | T    | F    | F                   | T                                                       |
| F | T | T                 | F    | T    | T                   | T                                                       |
| F | F | T                 | T    | T    | T                   | T                                                       |

### Contradiction (All false) :-

If a statement gives all values false in the last column, it is called a contradiction.

### Contingency (True / False) :-

If a statement gives all values true and false (both) in the last column, it is called contingency.

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Q: From the following formulas find out tautology, contingency and contradiction.

1.  $p \rightarrow p \wedge (p \vee q)$

2.  $(p \wedge \neg q) \vee (\neg p \wedge q)$

3.  $\neg(p \vee q) \vee (\neg p \vee \neg q)$

4.  $p \rightarrow (q \rightarrow p)$

5.  $p \vee \neg(p \wedge q)$

1.

| p | q | $p \vee q$ | $p \wedge (p \vee q)$ |
|---|---|------------|-----------------------|
| T | T | T          | T                     |
| T | F | T          | T                     |
| F | T | T          | F                     |
| F | F | F          | F                     |

Therefore, the above statement is contingency.

2.

| p | q | $\neg p$ | $\neg q$ | $p \wedge \neg q$ | $\neg p \wedge q$ | $(p \wedge \neg q) \vee (\neg p \wedge q)$ |
|---|---|----------|----------|-------------------|-------------------|--------------------------------------------|
| T | T | F        | F        | F                 | F                 | F                                          |
| T | F | F        | T        | T                 | F                 | T                                          |
| F | T | T        | F        | F                 | T                 | T                                          |
| F | F | T        | T        | F                 | F                 | F                                          |

Therefore, the above statement is contingency.



3.

| $P$ | $Q$ | $\neg P$ | $\neg Q$ | $P \vee Q$ | $\neg(P \vee Q)$ | $(\neg P \vee \neg Q)$ | $\neg(P \vee Q) \vee (\neg P \vee \neg Q)$ |
|-----|-----|----------|----------|------------|------------------|------------------------|--------------------------------------------|
| T   | T   | F        | F        | T          | F                | F                      | F                                          |
| T   | F   | F        | T        | T          | F                | T                      | T                                          |
| F   | T   | T        | F        | T          | F                | T                      | T                                          |
| F   | F   | T        | T        | F          | T                | T                      | T                                          |

Therefore, the given statement is a contingency.

4.

| $P$ | $Q$ | $Q \rightarrow P$ | $P \rightarrow (Q \rightarrow P)$ |
|-----|-----|-------------------|-----------------------------------|
| T   | T   | T                 | T                                 |
| T   | F   | T                 | T                                 |
| F   | T   | T                 | T                                 |
| F   | F   | T                 | T                                 |

Therefore, the given statement is a ~~contingency~~ tautology.

5.

| $P$ | $Q$ | $P \wedge Q$ | $\neg(P \wedge Q)$ | $P \vee \neg(P \wedge Q)$ |
|-----|-----|--------------|--------------------|---------------------------|
| T   | T   | T            | F                  | T                         |
| T   | F   | F            | T                  | T                         |
| F   | T   | F            | T                  | T                         |
| F   | F   | F            | T                  | T                         |

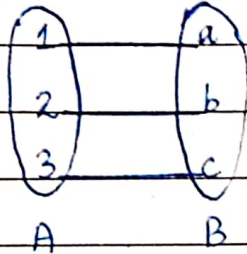
Therefore, the given statement is a ~~contingency~~ tautology.



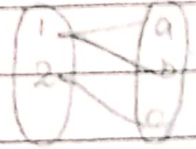
Lecture 3

Functions: A function is a relation b/w sets such that each element of A must be related to B.

$$f: A \rightarrow B$$



(Domain) [Range (Co-Domain)]



Not a function



\* All domains must have at least single range.

\* No domain can either have 2 ranges or no range.

Example

~~Example~~ Application:-

$$A = \{2, 3, 4\}^m \text{ } B = \{5, 6\}^n$$

Q: Determine all the functions from A to B.

$$A = 11^m$$

$$A = 2^3 = 8$$

$$(i) \{(2,5)(3,5)(4,5)\} \quad 1$$

$$(ii) \{(2,6)(3,6)(4,6)\} \quad 1$$

$$(iii) \{(2,5)(3,5)(4,6)\} \quad 2$$

$$(iv) \{(2,5)(3,6)(4,6)\} \quad 3$$

$$(v) \{(2,6)(3,5)(4,5)\} \quad 6$$

$$(vi) \{(2,6)(3,6)(4,5)\} \quad 5$$

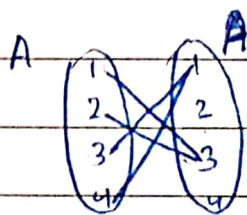
$$(vii) \{(2,6)(3,5)(4,6)\} \quad 7$$

$$(viii) \{(2,5)(3,6)(4,5)\} \quad 8$$

Example :-

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Q:  $A = \{1, 2, 3, 4\}$  & the function  $f: A \rightarrow A$  as shown in fig.



The image of each element of  $A$  :-

\*  $f(1) = 3$

\*  $f(2) = 3$

\*  $f(3) = 1$

\*  $f(4) = 1$

The image of  $f(A)$  of the function :-

$f(A) = \{1, 3\}$

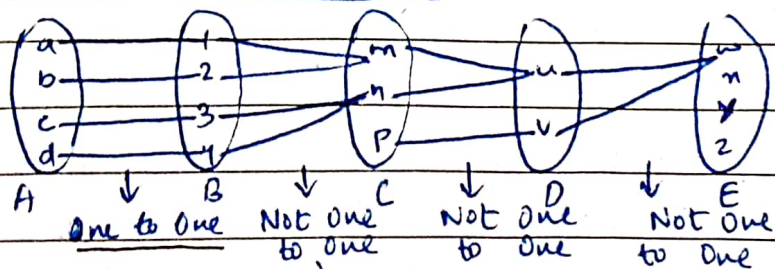
Types of functions :-

(i) One to One (injective)

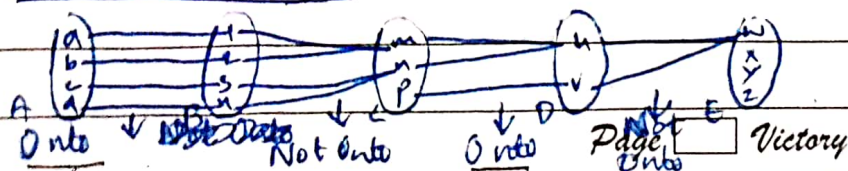
(ii) Onto (surjective)

(iii) <sup>or invertible</sup> One to One and Onto (bijective)

One to One (injective) :-

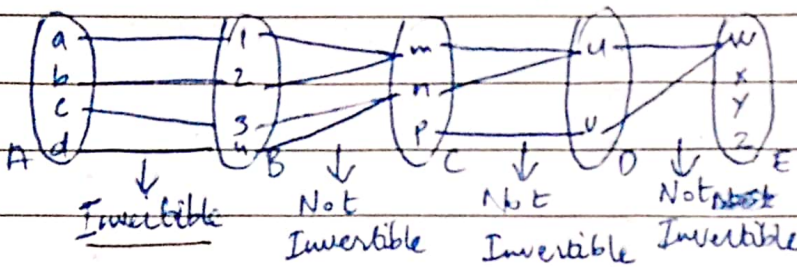


Onto (surjective) :-



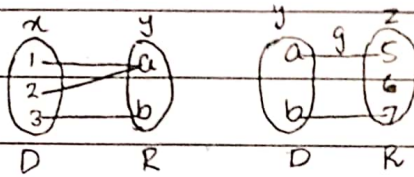


Invertible (bijective) :-



Composition of function :-

Example :-



$$(g \circ f)(1) = g[f(1)] = g(a) = 5$$

$$(g \circ f)(2) = g[f(2)] = g(a) = 5$$

$$(g \circ f)(3) = g[f(3)] = g(b) = 7$$

Example :-

Find  $f \circ g$  &  $g \circ f$  where  $f(x) = \sqrt{4+x^2}$  &  $g(x) = x + \frac{1}{4}$

$$\Rightarrow (g \circ f)(x) = g[f(x)] = g(\sqrt{4+x^2}) = \sqrt{4+x^2} + \frac{1}{4}$$

$$\Rightarrow (f \circ g)(x) = f[g(x)] = f\left(x + \frac{1}{4}\right) = \sqrt{4 + \left(x + \frac{1}{4}\right)^2}$$

Example :-

$$f(x) = x+2, \quad g(x) = \frac{1}{x+3}$$

$$(i) (f \circ g)(1) = f[g(1)] = f\left(\frac{1}{4}\right) = \frac{1}{4} + 2 = \frac{9}{4}$$

$$(ii) (g \circ f)(1) = g[f(1)] = g(3) = \frac{1}{3+3} = \frac{1}{6}$$

$$(iii) (f \circ g)(x) = f[g(x)] = f\left(\frac{1}{x+3}\right) = \frac{1}{x+3} + 2$$

$$(iv) (g \circ f)(x) = g[f(x)] = g(x+2) = \frac{1}{x+2+3} = \frac{1}{x+5}$$

$f(9(2))$

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$$(v) \log(2) = \frac{11}{7}$$

$$(vi) \log(2) = \frac{1}{7}$$

lecture 4

18/10/2023

Floor and Ceiling function :-

Floor  $\lfloor x \rfloor$

Ceiling  $\lceil x \rceil$

$$x = 3.2$$

$$\lfloor 3.2 \rfloor = 3$$

$$\lceil 3.2 \rceil = 4$$

$$x = 7$$

$$\lfloor 7 \rfloor = 7$$

$$\lceil 7 \rceil = 7$$

$$x = -5.2$$

$$\lfloor -5.2 \rfloor = -6$$

$$\lceil -5.2 \rceil = -5$$

example :-

$$\Rightarrow \left\lceil \frac{1}{2} + \left\lfloor \frac{3}{2} + \frac{1}{4} \right\rfloor + \left\lceil \frac{3}{2} + \frac{1}{5} \right\rceil \right\rceil$$

$$\Rightarrow \left\lceil \frac{1}{2} + \left\lfloor \frac{7}{4} \right\rfloor + \left\lceil \frac{11}{10} \right\rceil \right\rceil$$

$$\Rightarrow \left\lceil 0.5 + \lfloor 1.75 \rfloor + \lceil 1.1 \rceil \right\rceil$$

$$\Rightarrow \lceil 0.5 + 1 + 2 \rceil$$

$$\Rightarrow \lceil 3.5 \rceil \Rightarrow 4 \quad \text{Ans}$$



Example 2:

$$\left\lfloor \frac{3}{2} + \lceil 1.7 \rceil + \lfloor 2.5 \rfloor \right\rfloor$$

$$\Rightarrow \left\lfloor \frac{3}{2} + 2 + 2 \right\rfloor$$

$$\Rightarrow \left\lfloor \frac{3}{2} + 4 \right\rfloor$$

$$\Rightarrow \left\lfloor \frac{11}{2} \right\rfloor$$

$$\Rightarrow \lfloor 5.5 \rfloor \Rightarrow 5 \text{ Ans}$$

Integer function:

$$\text{INT}(3.14) = 3$$

$$\text{INT}(\sqrt{5}) = 2$$

$$\text{INT}(7) = 7$$

$$\text{INT}(-8.6) = -8$$

Absolute Value function :-

$$f(x) = x \text{ if } x \geq 0$$

$$f(x) = -x \text{ if } x < 0$$

$$|-15| = 15$$

$$|15| = 15$$

$$|-4.444| = 4.444$$

Logarithmic Function :-

$$y = \log_b x$$

$$b^y = x$$

$$\log_2 8 = 3$$

$$2^3 = 8$$

$$\log_{10} 300 = 2.4771$$

Exponential Function :-

$$a^m = a \cdot a \cdot a \dots a \text{ (m times)}$$

$$a^0 = 1$$

$$a^{-m} = \frac{1}{a^m}$$

$$a^{m/n} = (a^m)^{1/n} = \sqrt[n]{a^m}$$

Factorial Function :-

$$0! = 1$$

$$1! = 1$$

$$2! = 2 \times 1 = 2$$

$$3! = 3 \times 2 \times 1 = 6$$

$$4! = 4 \times 3 \times 2 \times 1 = 24$$

$$\vdots$$

$$n! = n(n-1)(n-2) \dots 1$$

Fibonacci Sequence :-

0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, ...

Recursively defined functions :-

Example :-

$$L(n) = \begin{cases} 0 & \text{if } n=1 \\ \lfloor L(\frac{n}{2}) \rfloor + 1 & \text{if } n>1 \end{cases}$$

Find  $L(25)$ .

$$L(25) = \lfloor L(\frac{25}{2}) \rfloor + 1 = L(12) + 1$$

$$L(12) = \lfloor L(\frac{12}{2}) \rfloor + 1 = L(6) + 1$$



$$L(6) = L\left[\frac{6}{2}\right] + 1 = L(3) + 1$$

$$L(3) = L\left[\frac{3}{2}\right] + 1 = L(1) + 1$$

$$L(1) = 0 \text{ } \cancel{= 0} = 0$$

$$L(25) = L(12) + 1$$

$$= L(6) + 2$$

$$= L(3) + 3$$

$$= L(1) + 4$$

$$= 0 + 4$$

$$\boxed{L(25) = 4}$$

Example:-

$$Q(a, b) = \begin{cases} 0 & \text{if } a < b \\ Q(a-b, b) + 1 & \text{if } a \geq b \end{cases}$$

$$Q(12, 3) = 0$$

$$Q(14, 3) = Q(11, 3) + 1$$

$$Q(11, 3) = Q(8, 3) + 2$$

$$Q(8, 3) = Q(5, 3) + 3$$

$$Q(5, 3) = Q(2, 3) + 4$$

$$= 0 + 4$$

$$\boxed{Q(14, 3) = 4}$$

Example 1:-

$f(n) = \mathbb{R} \rightarrow \mathbb{R}$  is defined by

$$f(n) = n^2 - 3n \quad \text{if } n \geq 2$$

$$f(n) = n + 2 \quad \text{if } n < 2$$

Find  $f(0)$ ,  $f(5)$ ,  $f(-1)$

$f(0)$  when  $x+2$  if  $x \leq 2$

$$f(0) = 0+2$$

$$\boxed{f(0) = 2}$$

$f(5)$  when  $x^2-3x$  if  $x \geq 2$

$$f(5) = (5)^2 - 3(5)$$

$$f(5) = 25 - 15$$

$$\boxed{f(5) = 10}$$

$f(-2)$  when  $x+2$  if  $x \leq 2$

$$f(-2) = -2+2$$

$$\boxed{f(-2) = 0}$$

Example 2 :-

Consider

$$f(x) = x^2 + 3x + 1$$

$$g(x) = 2x - 3$$

$$g \circ f = 2(x^2 + 3x + 1) - 3$$

$$g \circ f = 2x^2 + 6x + 2 - 3$$

$$\boxed{g \circ f = 2x^2 + 6x - 1}$$

$$f \circ g = (2x-3)^2 + 3(2x-3) + 1$$

$$f \circ g = (2x)^2 - 2(2x)(3) + (3)^2 + 6x - 9 + 1$$

$$f \circ g = 4x^2 - 12x + 9 + 6x - 9 + 1$$

$$f \circ g = 4x^2 - 12x + 6x + 1$$

$$\boxed{f \circ g = 4x^2 - 6x + 1}$$

$$g \circ f(2) = 2[(2)^2 + 3(2) + 1] - 3$$



$$\Rightarrow g \circ f(2) = 2(4+6+1)-3$$

$$\Rightarrow g \circ f(2) = 2(11)-3$$

$$\Rightarrow g \circ f(2) = 22-3$$

$$\Rightarrow \boxed{g \circ f(2) = 19}$$

$$f \circ g(3) = [2(3)-3]^2 + 3[2(3)-3] + 1$$

$$f \circ g(3) = [6-3]^2 + 3[6-3] + 1$$

$$f \circ g(3) = 3^2 + 3(3) + 1$$

$$f \circ g(3) = 9 + 9 + 1$$

$$\boxed{f \circ g(3) = 19}$$

Example 3 :-

$$f(x) = 3x-1 \text{ if } x > 3$$

$$f(x) = x^2-2 \text{ if } -2 \leq x \leq 3$$

$$f(x) = 2x+3 \text{ if } x < -2$$

find  $f(2)$ ,  $f(4)$ ,  $f(-1)$ ,  $f(-3)$

$$f(2) \text{ on } f(x) = \cancel{x^2-2} x^2-2 \text{ if } -2 \leq x \leq 3$$

$$\Rightarrow (2)^2-2$$

$$\Rightarrow 4-2$$

$$\Rightarrow \boxed{f(2) = 2}$$

$$f(4) \text{ on } f(x) = 3x-1 \text{ if } x > 3$$

$$\Rightarrow f(4) = 3(4)-1$$

$$\Rightarrow f(4) = 12-1$$

$$\Rightarrow \boxed{f(4) = 11}$$

$$f(-1) \text{ on } f(x) = x^2 - 2 \text{ if } -2 \leq x \leq 3$$

$$\Rightarrow f(-1) = (-1)^2 - 2$$

$$\Rightarrow f(-1) = 1 - 2$$

$$\Rightarrow \boxed{f(-1) = -1}$$

$$f(-3) \text{ on } f(x) = 2x + 3 \text{ if } x < -2$$

$$\Rightarrow f(-3) = 2(-3) + 3$$

$$\Rightarrow f(-3) = -6 + 3$$

$$\Rightarrow \boxed{f(-3) = -3}$$

## Lecture 5

24/10/2023

Relations:

$$A = \{a, b, c\}, B = \{x, s, t\}$$

$$R = \{(a, x)(b, x)(b, t)(c, s)\} \text{ is a relation.}$$

\* Every function has a relation but every relation does not have a function.

To calculate total relation.

$$\Rightarrow 2^{m \times n}$$

$$\Rightarrow 2^{3 \times 3}$$

$$\Rightarrow 2^9$$

$$\Rightarrow 512$$

Example :-

$$A = \{1, 2\}$$

$$\Rightarrow 2^m = 2^2 = 4$$

$$A_1 = (1, 1)$$

$$A_2 = (1, 2)$$

$$A_3 = (2, 1)$$

$$A_4 = (2, 2)$$



Complement:-

$$X = \{1, 2, 3\}, Y = \{8, 9\}$$

$$X \times Y = \{(1, 8)(1, 9)(2, 8)(2, 9)(3, 8)(3, 9)\}$$

$$\text{Given } R = \{(1, 8)(2, 8)(1, 9)(3, 9)\}$$

- find complement of given R.

$$\bar{R} = \{(2, 9)(3, 8)\}$$

Inverse:-

$$R = \{(2, a)(2, c)(3, b)(3, c)(4, a)\}$$

$$R^{-1} = \{(a, 2)(c, 2)(b, 3)(c, 3)(a, 4)\}$$

\* Here, inverse means orders are changed i.e. domain change into range, range change into domain.

Relation as matrix:-

$$P = \{1, 2, 3, 4\}, Q = \{a, b, c, d\}$$

$$R = \{(1, a)(1, b)(1, c)(2, b)(2, c)(2, d)(3, a)(3, d)(4, b)(4, c)\}$$

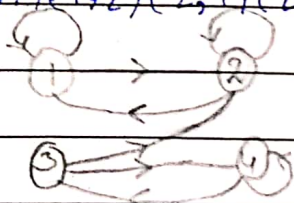
|        |   | a | b | c | d |
|--------|---|---|---|---|---|
| Domain | 1 | 1 | 1 | 1 | 0 |
|        | 2 | 0 | 1 | 1 | 1 |
|        | 3 | 1 | 0 | 0 | 1 |
|        | 4 | 0 | 1 | 1 | 0 |

Column Domain

Relation as a directed graph:-

$$P = \{1, 2, 3, 4\}$$

$$R = \{(1, 1)(1, 2)(2, 1)(2, 2)(3, 2)(3, 4)(4, 3)(4, 4)(4, 2)\}$$

Relation as an arrow diagram:-

$$P = \{1, 2, 3, 4\}$$

$$Q = \{a, b, c, d\}$$

$$R = \{(1, a)(2, a)(3, a)(1, b)(4, b)(4, c)(4, d)\}$$



Properties:-1) Reflexive

\*  $(a, a) \in R$

$A = \{1, 2, 3\}$

$R = \{(1,1)(2,2)(1,3)(1,2)(2,3)(3,3)\}$

The relation is reflexive.

2) Symmetric

\*  $(a, b) \in R$  then  $(b, a) \in R$ .

$A = \{1, 2, 3\}$

$R = \{(1,1)(1,2)(2,1)(1,3)(3,1)(3,2)(2,2)(2,3)\}$

 $R$  is symmetric.3) Transitive

\*  $(a, b) \in R$  &  $(b, c) \in R$  then  $(a, c) \in R$

$A = \{1, 2, 3\}$

$R = \{(1,2)(2,2)(2,1)(1,1)\}$

 $R$  is transitive.Example

Let  $A = \{4, 5, 6\}$

$R = \{(3,4)(4,3)(5,4)(5,3)(3,3)\}$

 $R$  is not transitive since  $(4,3) \in R$  &  $(3,4) \in R$  but  $(4,4) \notin R$ .4) Anti-Symmetric

$(a, b) \in R \rightarrow (b, a) \notin R$  Unless  $a = b$

Example:-

$R = \{(4,4)(4,5)(5,6)(4,6)\}$

 $R$  is antisymmetric.Example:-

$R = \{(4,4)(4,5)(5,6)(6,5)(4,6)\}$

 $R$  is not antisymmetric.



Lecture #6

Equivalence Relation

Partial order Relation

Reflexive

Reflexive

Symmetric

Anti-symmetric

Transitive

Transitive

Example :-

$$\text{let } A = \{1, 2, 3, 4\}$$

$$R = \{(1,1)(1,3)(2,2)(2,4)(3,1)(3,3)(4,2)(4,4)\}$$

R is equivalence R?

1) Reflexive ✓

2) Symmetric ✓

3) Transitive ✓

⇒ Equivalence R

Example :-

$$A = \{0, 1, 2, 3\}$$

$$R = \{(0,0)(1,1)(1,2)(2,3)(2,2)(3,2)(3,3)(1,0)(3,0)(3,1)(2,1)(1,3)(0,2)(0,3)(2,0)\}$$

1) Reflexive ✓

2) Symmetric x  $(1,0) \in R$  but  $(0,1) \notin R$ 

⇒ Not equivalence

Q: Show that the relation  $(x, y) \in R$  if  $x \geq y$  defined on the set of positive integers is a partial order relation.

Sol:-

Let  $A = \{1, 2, 3, 4\}$  and  $x \geq y$  then

$$R = \{(1,1)(2,2)(3,3)(4,4)(2,1)(3,1)(4,1)(3,2)(4,2)(4,3)\}$$

1) Reflexive ✓

2) Antisymmetric ✓

3) Transitive ✓

$\Rightarrow$  Partial order relation

Q: Show that the relation  $(x, y) \in R$  if  $x \leq y$  defined on the set of +ve integers is a partial order relation.

Sol:-

Let  $A = \{1, 2, 3, 4\}$  and  $x \leq y$  then

$$R = \{(1,1)(2,2)(3,3)(4,4)(1,2)(1,3)(1,4)(2,3)(2,4)(3,4)\}$$

1) Reflexive ✓

2) Antisymmetric ✓

3) Transitive ✓

$\Rightarrow$  Partial order relation

Principle of Inclusion & Exclusion:-

First Principle: If  $P$  and  $Q$  are disjoint

$$|P \cup Q| = |P| + |Q|$$

Theorem 1: Let  $P$  and  $Q$  be any two non-disjoint sets

$$|P \cup Q| = |P| + |Q| - |P \cap Q|$$



Theorem 2: If  $P, Q$  &  $R$  are three finite set.

$$|P \cup Q \cup R| = |P| + |Q| + |R| - |P \cap Q| - |P \cap R| - |Q \cap R| + |P \cap Q \cap R|$$

Example:-

Among the first 500 +ve integers. Determine the integers that are not divisible by 2 nor by 3 nor by 5.

Sol:-

Let  $A$  be the number of integers divisible by 2.

Let  $B$  be the number of integers divisible by 3.

Let  $C$  be the number of integers divisible by 5.

$$\Rightarrow |A| = \left\lfloor \frac{500}{2} \right\rfloor = 250$$

$$|A \cap B| = \left\lfloor \frac{500}{2 \times 3} \right\rfloor = 83$$

$$\Rightarrow |B| = \left\lfloor \frac{500}{3} \right\rfloor = 166$$

$$|A \cap C| = \left\lfloor \frac{500}{2 \times 5} \right\rfloor = 50$$

$$\Rightarrow |C| = \left\lfloor \frac{500}{5} \right\rfloor = 100$$

$$|B \cap C| = \left\lfloor \frac{500}{3 \times 5} \right\rfloor = 33$$

$$|A \cap B \cap C| = \left\lfloor \frac{500}{2 \times 3 \times 5} \right\rfloor = 16$$

$$\Rightarrow |A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

$$\Rightarrow |A \cup B \cup C| = 250 + 166 + 100 - 83 - 50 - 33 + 16$$

$$\Rightarrow |A \cup B \cup C| = 366$$

$$\Rightarrow 500 - 366$$

$$\Rightarrow 134 \text{ (Not divisible by 2, 3 or 5)}$$

## Lecture #7

Date: 07/10/2023

### Pigeonhole Principle

$$\text{Formula} = \left\lceil \frac{N}{k} \right\rceil \rightarrow \begin{array}{l} \text{object} \\ \text{boxes} \end{array}$$

Example: Show that if there are 30 students in a class then at least two have last name that begin with the same letter.

$$\Rightarrow \left\lceil \frac{N}{k} \right\rceil = \left\lceil \frac{30}{26} \right\rceil = \left\lceil 1.15 \right\rceil = 2$$

### Counting Techniques